

**Advanced Programming 2025-2026 Assignment #3: Deadline: 22/5/2026**  
**Second semester Dr. Ala Abuthawabeh and Dr. Ali Azawi**

**perform the following Matrix Operations:**

- 1- using only in Python (Write four functions):**
- 2- using Numpy library built in functions**

- Matrix Addition  $\mathbf{A} + \mathbf{B}$
- Matrix Multiplication  $\mathbf{A} \cdot \mathbf{B}$
- Determinant Calculation  $\det(\mathbf{A})$
- Matrix Inverse  $\mathbf{A}^{-1}$
- Solve Systems of Linear Equations

Instructions: submit your files via Moodle elearning. (you can use your github url)

**Remark: Copy Paste is not allowed. And any two assignments with a similar solution, zero marks will be given for both.**